

Sidhant Yadav

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WORK EXPERIENCE

LEAD SOFTWARE DEVELOPER (ModivaStudios)

Jan 2026 - Apr 2026

- Architected and developed a full-stack AI-powered video generation platform clawdvideo.com using **React (JSX)**, **Vite**, and **Node.js/ Express**, delivering a production-grade web application from scratch for a startup in the **Google Accelerator**.
- Engineered a real-time AI agent pipeline integrating **Google's Veo** generative video models via a custom **RESTful API** layer, enabling prompt-to-video synthesis with server-side orchestration through a dedicated agent architecture (agent.js)
- Built a modular, component-based frontend with **Tailwind CSS** and custom **CSS transitions**, implementing distinct layout patterns (full-width cinematic reveals, editorial asymmetric grids, accordion stacks) with full responsiveness across devices.
- Integrated **Clerk** authentication for secure user session management, coordinating a **multi-branch Git workflow** with feature branches, merge conflict resolution, and coordinated releases to a shared upstream repository
- Led end-to-end UI/UX development, translating product vision into interactive prototypes and iterating on design systems including typography hierarchies, glass morphism card components, ambient glow effects, and micro-animations

WEBSITE FREELANCING COMPANY FOUNDER (NovaSolutions .Inc)

June 2022 - Sept 2025

- Founded and managed an independent web development venture, proactively reaching out to local small businesses to design and build customized websites that showcased their goods, services, and brand identity in a professional digital format.
- Oversaw a team of **five junior coders**, delegating tasks according to skill level and project deadlines, while mentoring team members on front end development using **HTML, CSS, JavaScript**. And back-end integration using **PHP, MySQL**.
- Took initiative in client acquisition by cold-calling and direct messaging potential customers, communicating the value of an online presence, preparing detailed pitches, and presenting mockup designs to secure contracts with multiple businesses.

PROJECTS

FINANCIAL NEWS ETL & DASHBOARD

- Built a production-grade ETL pipeline using **Prefect** to automatically extract daily financial news from global REST APIs — fully hands-off after deployment
- Transformed raw JSON data with **Pandas** and ran real-time sentiment analysis on market headlines using **NLTK (VADER)**, producing clean, analysis-ready records
- Designed a relational database schema with **SQLAlchemy** for structured data ingestion, and containerized the entire backend with **Docker** for portable, cloud-agnostic deployment
- Developed a full-stack analytics dashboard using **Next.js / React** with a glassmorphism UI, displaying live sentiment trends and macroeconomic insights pulled directly from the database

ENTERPRISE DOCUMENT INTELLIGENCE SYSTEM (RAG PIPELINE)

- Engineered a full-stack Retrieval-Augmented Generation (RAG) application that instantly processes large-scale PDF documents, transforming unstructured text into an interactive knowledge base.
- Architected a high-performance vector search engine using **LangChain** and **ChromaDB**, achieving lightning-fast, sub-second query responses across hundreds of pages of dense documentation.
- Developed an intuitive, real-time chat interface using **Streamlit** that delivers highly accurate, citation-backed answers, completely eliminating AI hallucinations and ensuring 100% data reliability for end users.

EMBEDDED TRAFFIC CONTROL SYSTEM

- Engineered an automated, real-time traffic intersection controller in C++, implementing Finite State Machine (FSM) architectures and deterministic timing algorithms to optimize logic flow.
- Architected custom hardware prototypes utilizing transistors, Integrated Circuits (ICs), and breadboards to bridge the gap between microcontroller software and physical electromechanical systems.
- Integrated hardware-software interfaces utilizing Pulse Width Modulation (PWM), digital/analog I/O, and low-level logic controls to process real-time sensor telemetry and drive sequential outputs.

EDUCATION

The University of Waterloo
Bachelor of Applied Science, Majoring in Computer Engineering

Waterloo, ON
2025- 2030